CLINICAL · EAR

Vestibular schwannoma and CP angle tumours

A vestibular schwannoma is a slow, benign tumour of the eighth nerve sheath that announces itself through the ear: one-sided hearing loss and tinnitus, with speech understanding worse than the audiogram predicts. An MRI makes the diagnosis; the choice to watch, irradiate or remove depends on size, hearing, age and growth.

AT A GLANCE

- Asymmetric sensorineural loss with poor speech understanding.
- Gadolinium MRI is the gold standard.
- Watch, radiate or remove by size, hearing and age.
- The route depends on tumour size and hearing.
- Bilateral tumours mean NF2.
- Most small tumours are watched (17–29% grow over 4 years); radiosurgery or surgery chosen by size and hearing.

1 CHAPTER ONE Overview

The ear is the first to know.

1.1 Definition

A benign tumour of Schwann cells, usually arising from the vestibular division of the eighth nerve in the internal auditory canal. *Acoustic neuroma* is the traditional, inaccurate name: it is rarely acoustic and never a neuroma.

1.2 Epidemiology

It accounts for 8 to 10% of intracranial tumours and about 80% of cerebellopontine angle tumours, usually presenting between 40 and 60. Most are sporadic and one-sided. Bilateral tumours mean *neurofibromatosis type 2*.

1.3 Why it matters

It is the commonest serious cause of asymmetric sensorineural hearing loss, and every asymmetric loss is screened for it. Untreated large tumours compress the brainstem and cause hydrocephalus.

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2 CHAPTER TWO Pathology

Slow growth from inside the canal.

2.1 Origin and growth

The tumour arises near the glial-Schwann cell junction in the internal auditory canal, widening the canal before it grows into the cerebellopontine angle. Growth is slow, around 1 to 2 mm a year, and some tumours never grow at all.

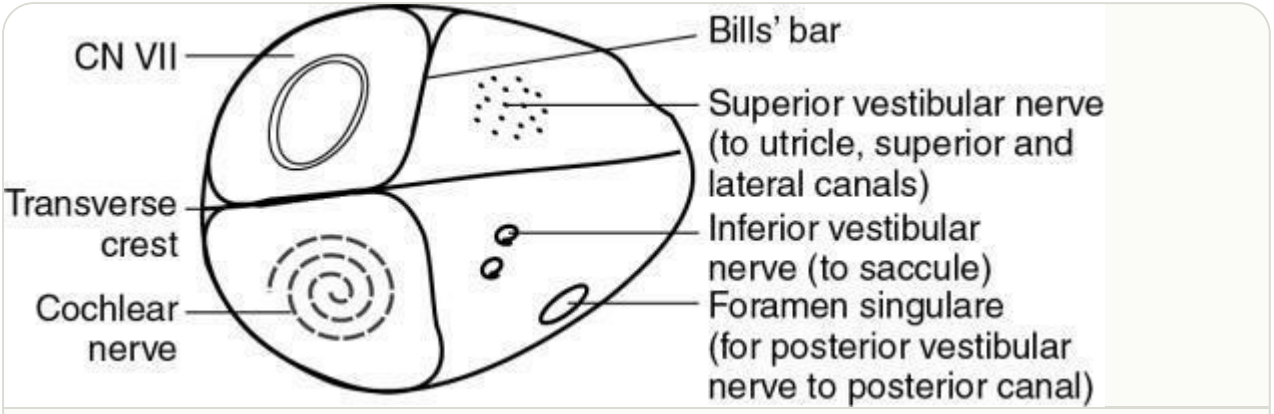


FIGURE 1 The fundus of the internal auditory canal, showing the position of the facial, cochlear and superior and inferior vestibular nerves.
Dhingra, fig. 18.1

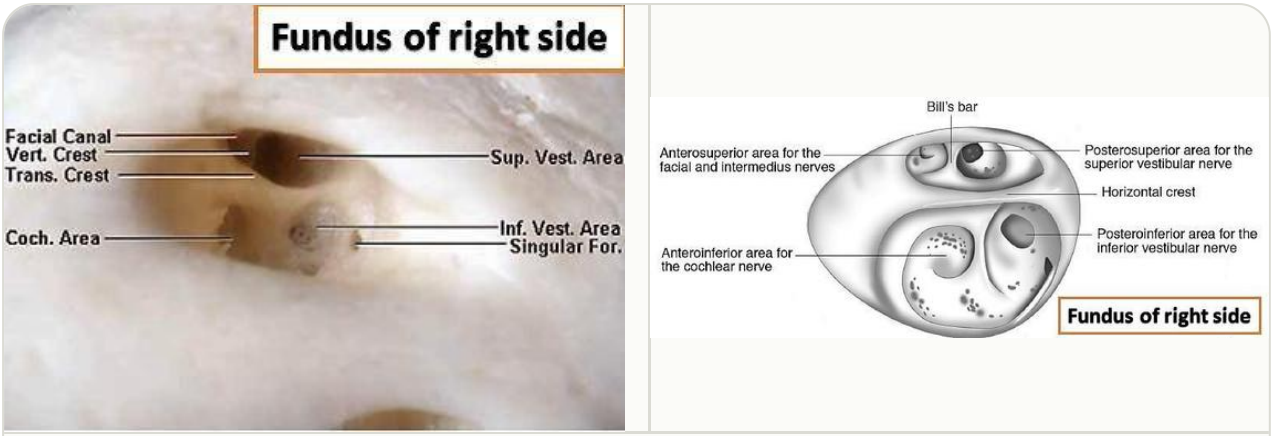


FIGURE 2 The fundus of the internal auditory canal, with Bill's bar and the transverse crest separating the facial, cochlear and vestibular nerves.

Anatomy atlas · ENT Steps p. 83

2.2 Histology

Two patterns: *Antoni A*, compact spindle cells with palisading nuclei (Verocay bodies), and *Antoni B*, loose, cystic tissue.

2.3 Neurofibromatosis type 2

Autosomal dominant, from loss of the NF2 gene on chromosome 22. Bilateral vestibular schwannomas, often with meningiomas and other schwannomas, present in young adults.

2.4 Size

SIZE AND SPREAD

STAGE	SIZE (DHINGRA)	WHAT IT AFFECTS
Intracanalicular	In the canal	Eighth nerve
Small	Up to 1.5 cm	Into the angle
Medium	1.5 to 4 cm	Fifth nerve, cerebellum
Large	Over 4 cm	Brainstem, fourth ventricle, hydrocephalus

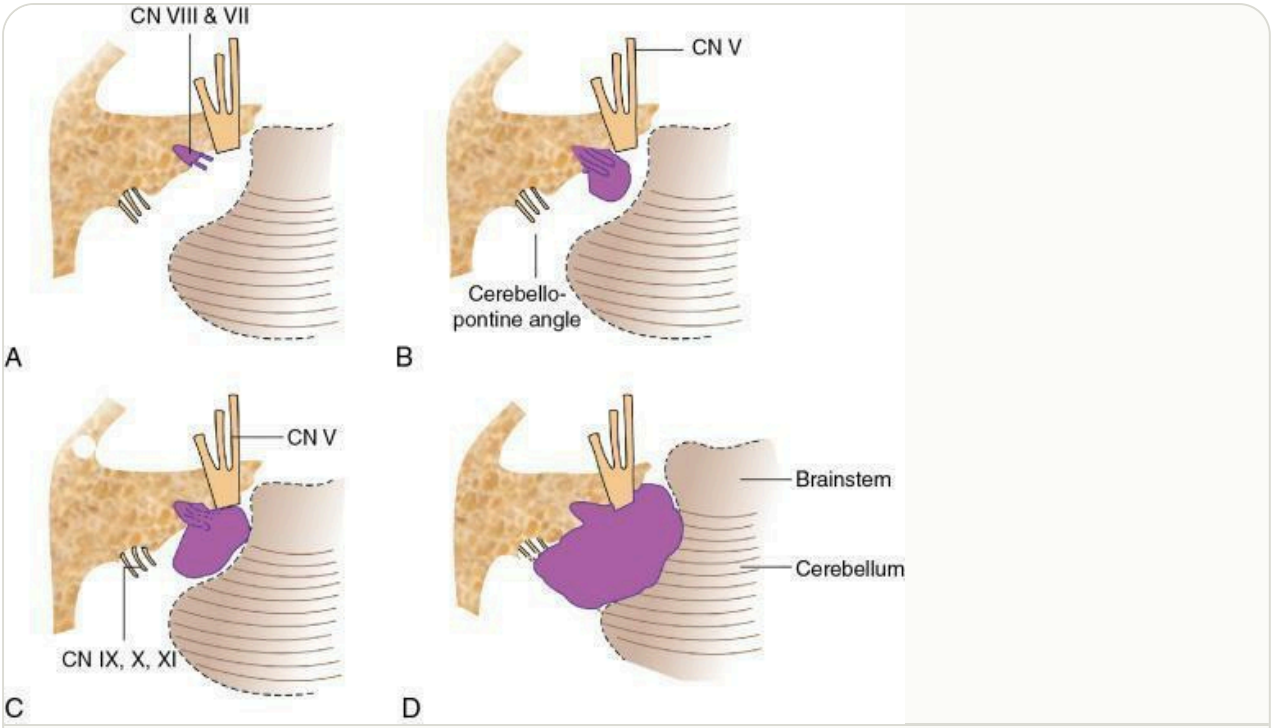


FIGURE 3 Expansion of an acoustic neuroma: (A) intracanalicular, (B) into the cerebellopontine angle, (C) pressing on the fifth nerve, (D) onwards to the brainstem and lower cranial nerves.
Dhingra, fig. 18.2

3 CHAPTER THREE Clinical features

A stepwise spread.

3.1 Eighth nerve

3.1.1 Hearing loss

Progressive, one-sided sensorineural loss with poor speech discrimination, often far worse than the pure-tone thresholds suggest. About one in ten presents as sudden hearing loss.

3.1.2 Tinnitus

One-sided and persistent.

3.1.3 Balance

Unsteadiness rather than true vertigo, because the brain compensates as slowly as the tumour grows.

3.2 Fifth nerve

At around 2.5 cm: reduced corneal reflex, often the earliest sign, then facial numbness.

3.3 Seventh nerve

The sensory fibres fail first: numbness of the posterior meatal wall (*Hitselberger's sign*), loss of taste and reduced lacrimation. Motor weakness of the face comes late.

3.4 Lower cranial nerves

Ninth and tenth: dysphagia, hoarseness and a reduced gag.

3.5 Cerebellum and brainstem

Ataxia, nystagmus, and finally raised intracranial pressure with headache, vomiting and papilloedema from obstructed CSF flow.

> **URGENT**
> Headache, vomiting, ataxia or papilloedema with a large tumour means hydrocephalus or brainstem compression: urgent neurosurgical review.

4 CHAPTER FOUR Investigations

MRI first.

4.1 Audiometry

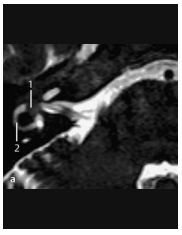
Asymmetric sensorineural loss, poor discrimination, *roll-over* (discrimination falls as loudness rises), absent or decaying stapedial reflex, and abnormal tone decay: the signs of a retrocochlear lesion.

4.2 MRI with gadolinium

The gold standard: finds tumours of a few millimetres. A brightly enhancing mass centred on the internal auditory canal, with a cone into the angle (*ice-cream cone*).

IMAGING IN THE APP 9 cases

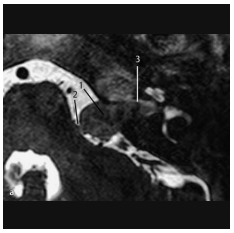
Intracanalicular and angle schwannomas, cystic change, radiotherapy effects, NF2, and the mimics: meningioma, epidermoid and arachnoid cyst.



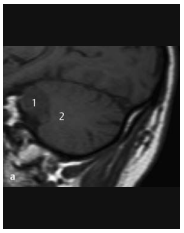
Vestibular Schwannoma
Tumours · 13 images



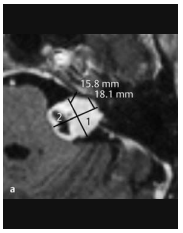
Schwannoma of the Internal Auditory Canal
Tumours · 3 images



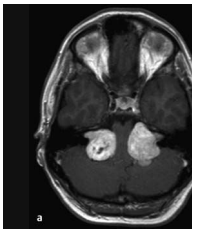
Schwannoma of the Cerebellopontine Angle
Tumours · 7 images



Cystic Degeneration in CPA Schwannoma
Tumours · 4 images



Effects of Radiotherapy on CPA Schwannoma
After surgery · 2 images



Neurofibromat II (NF II)
Tumours · 6 imag

4.3 Auditory brainstem response

Used when MRI is not possible. A prolonged inter-aural wave V latency or I–V interval suggests a lesion, but small tumours are missed.

WHERE THE BOOKS DIFFER · ABNORMAL WAVE V DIFFERENCE

Dhingra	Lee
More than 0.2 ms.	0.4 to 0.5 ms.

4.4 Caloric test

Reduced on the affected side in most, normal in very small tumours.

4.5 CT

If MRI cannot be done, and to show the bony canal and approach anatomy.

4.6 Avoid lumbar puncture

CSF protein is raised, but a puncture with a large tumour risks herniation.

4.7 Differential diagnosis

OTHER CEREBELLOPONTINE ANGLE LESIONS

LESION	DISTINGUISHING FEATURES
Meningioma	Broad dural base, dural tail, eccentric to the canal, calcification, hyperostosis
Epidermoid	Does not enhance; bright on diffusion-weighted MRI
Facial schwannoma	Facial palsy, enlarges the geniculate or labyrinthine segment
Lipoma	Bright on T1 before contrast; suppresses with fat saturation
Haemangioma	Ground-glass erosion; palsy out of proportion to size
Arachnoid cyst	Follows CSF on all sequences

5 CHAPTER FIVE Management

Watch, radiate, or remove.

5.1 Choosing

The decision rests on tumour size and growth, hearing in both ears, age and health, and the patient's wishes.

WHERE THE BOOKS DIFFER · FIRST CHOICE

Dhingra	Lee
Favours surgical removal.	Presents observation, radiosurgery and microsurgery as equal options, chosen by patient and tumour.

Modern practice follows Lee: many small tumours are simply watched.

5.2 Observation

For small tumours, older or unfit patients, and those with good hearing: serial MRI at 6 to 12 months, then less often. Treat if it grows.

5.3 Stereotactic radiosurgery

A single high dose (around 12 to 13 Gy at the margin), usually for tumours under 2.5 to 3 cm. Growth is controlled in 90 to 95%. Hearing is not reliably preserved; the facial nerve risk is about 1%; malignant change is very rare, about 1 in 1000. Conventional external beam radiotherapy has no role.

5.4 Microsurgery

SURGICAL APPROACHES

APPROACH	BEST FOR	HEARING	MAIN RISK
Translabyrinthine	Any size	Lost	CSF leak
Retrosigmoid (suboccipital)	Medial, small or large tumours	May be preserved	Headache, cerebellar retraction
Middle fossa	Canal tumours under 1.5 cm	Preserved in 60–70%	Temporal lobe retraction, facial nerve

5.4.1 Translabyrinthine

Through the mastoid and labyrinth: no brain retraction and the best view of the facial nerve, but hearing is sacrificed.

5.4.2 Retrosigmoid

Through the posterior fossa behind the sigmoid sinus: hearing can be preserved for tumours that spare the lateral canal; postoperative headache is commoner.

5.4.3 Middle fossa

Above the temporal bone: for small intracanalicular tumours with good hearing. The facial nerve lies over the tumour and is at greater risk.

OPERATIVE GUIDES IN THEATRE 3 operations

Acoustic Neuroma
Acoustic neuroma surgery

Stereotactic Radiosurgery
Stereotactic radiosurgery

Bone-Anchored Hearing Devices
Rehabilitating single-sided deafness

5.5 Subtotal removal

Leaving a little tumour on the facial nerve preserves facial function; the remnant often stays stable and can be watched or irradiated.

5.6 NF2

Bevacizumab may shrink tumours and improve hearing. An auditory brainstem implant or cochlear implant can restore some hearing.

6 CHAPTER SIX Complications

Of the tumour and of treatment.

- Facial palsy.
- Hearing loss.
- CSF leak and meningitis.
- Chronic headache, especially after the retrosigmoid route.
- Hydrocephalus and brainstem compression with large tumours.

CHOOSING BETWEEN WAITING, RADIOSURGERY AND SURGERY (OXFORD TEXTBOOK)

Observation first is now common for small tumours. In a large series followed for four years, only 17% of intracanalicular and 29% of extrameatal tumours grew, and growing tumours usually enlarge by 1 to 2 mm a year, so waiting rarely makes later treatment harder. The first repeat MRI is at six months, to catch a fast-growing tumour or a wrong diagnosis, then yearly while stable. The cost of waiting is hearing, which may decline even when the tumour does not grow.

Stereotactic radiosurgery, usually a single session of 11 to 14 Gy at the margin, controls most small and medium tumours. Surgery after failed radiosurgery carries a worse facial nerve outcome than primary surgery.

Surgical approach follows size and hearing. The *middle fossa* approach suits intracanalicular tumours up to about 1.5 cm with serviceable hearing. The *retrosigmoid* approach suits tumours under about 2 cm, not filling more than two-thirds of the internal meatus, with serviceable hearing; in a large review, class A or B hearing was kept in 57% of intracanalicular tumours, 37% under 2 cm and 12% over 2 cm. The *translabyrinthine* approach sacrifices hearing but gives direct access without brain retraction and can remove a tumour of any size; facial nerve outcomes at House–Brackmann I or II are high for tumours under about 3.5 cm. Guidelines find no approach clearly superior for facial function. Presenting features are mostly asymmetric hearing loss and tinnitus; facial numbness occurs in 7 to 9% and facial weakness in only about 5%, even with large tumours.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Origin:** the vestibular nerve in the internal auditory canal.
- **Presenting symptoms,** and why vertigo is uncommon.
- **Order of cranial nerve involvement:** VIII, V (corneal reflex), VII (Hitselberger's sign), IX and X.
- **Retrocochlear audiology:** discrimination, roll-over, reflex decay.
- **MRI with gadolinium** as the gold standard.
- **The three surgical approaches,** with hearing outcome for each.

- **NF2:** bilateral tumours, chromosome 22.

7.2 Common pitfalls

COMMON PITFALLS

- Not requesting MRI for an asymmetric loss.
- Attributing sudden hearing loss to a virus without imaging.
- Doing a lumbar puncture with a large tumour.
- Promising hearing preservation after radiosurgery.

Written for this app from Oxford Textbook of Otolaryngology ch. 18, Oxford Specialist Handbook ch. 19, Oxford Handbook of ENT ch. 8, Dhingra ch. 18 and K. J. Lee ch. 24. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

INNER EAR & IAC · TUMOURS

Vestibular Schwannoma

IMAGING FINDINGS

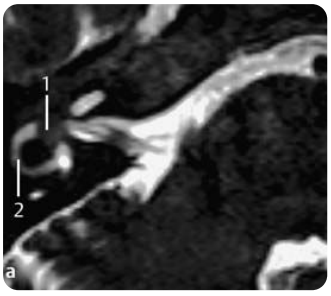
- Enhancing mass centred on the porus, "ice cream on a cone"
- Dark filling defect in bright CSF on high-resolution T2; fundal cap of fluid
- Acute angle with the petrous bone; widened porus; cystic change in larger tumours

TOP DIFFERENTIALS

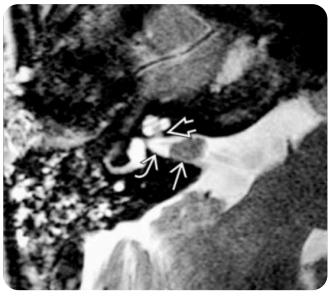
- Meningioma (broad dural base, dural tail)
- Epidermoid cyst (DWI bright)
- Facial nerve schwannoma (labyrinthine segment)

CHECKLIST

- Size in canal and cistern; brainstem and fourth ventricle (Koos)
- Fundal cap and cochlear nerve canal for hearing preservation



MRI, T2-weighted, axial.A partial lack of fluid in the vestibule and horizontal semicircular canals is evident (1). A mass lesion is present in the vestibul…



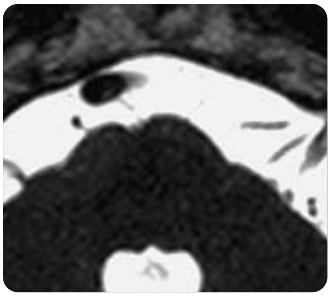
Axial T2WI MR reveals a small intracanalicular vestibular schwannoma visualized as a soft tissue intensity mass surrounded by…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1222.

INNER EAR & IAC · TUMOURS

Schwannoma of the Internal Auditory Canal

- Vestibular schwannoma is the most frequent pathology (90%) found in this area. The clinical presentation facilitates the diagnosis because of hearing problems (vestibulocochlear).
- Multiple cranial nerve deficits are suggestive of inflammatory processes.
- A meningioma with the internal auditory canal as its only location and site of origin is possible but extremely rare; these tumors are usually CPA lesions protruding into the canal. In cases of a suspicious lesion in the CPA, MRI with contrast enhancement is recommended.

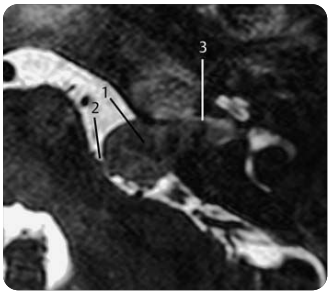


MRI, T2-weighted, axial. On this T2-weighted image, the cerebrospinal fluid has very high signal intensity. Normally, the internal auditory canal (…

INNER EAR & IAC · TUMOURS

Schwannoma of the Cerebellopontine Angle

- A vestibular schwannoma is the most frequent pathology (90 %) in this area. Its initial clinical presentation will help to determine the origin of the lesion, that is of hearing problems (vestibulocochlear) or facial paresis.
- Meningioma may show the so-called dural tail, although this is not always present, and involvement of the internal auditory canal is less common.
- Epidermoid cysts have a more rounded appearance, with contents which may also be hyperintense on T1-weighted images, but show lack of enhancement after contrast administration.
- Arachnoid cysts in the CPA may be confused with cystic tumors.
- Anterior localization may also indicate a trigeminal neurinoma.

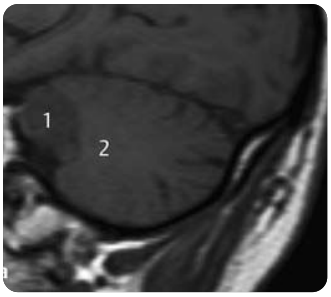


T2-weighted MRI, axial. Lesion of moderate size (1), partly within the internal auditory canal, and partly in the CPA, just touching the brainstem…

INNER EAR & IAC · TUMOURS

Cystic Degeneration in CPA Schwannoma

- An arachnoid cyst may be secondary to folding or inclusion of arachnoid membranes by compression from a schwannoma or meningioma. They are mostly located outside the margin of the causative lesion.
- A mucocele may be considered when the cyst is in located in a structure covered by mucosa.
- Although a hemangioblastoma only rarely presents as a CPA lesion, it may undergo rapidly progressive and massive cystic degeneration with partial enhancement of its solid parts, as is frequently observed in von Hippel–Lindau disease.



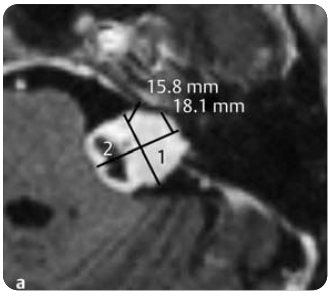
Patient with a vestibular schwannoma evaluated by a wait-and-see policy.

Dunnebier, Imaging for Otolaryngologists, Fig. 5.11 a

INNER EAR & IAC · AFTER SURGERY

Effects of Radiotherapy on CPA Schwannoma

- After stereotactic radiosurgery or fractionated radiotherapy, schwannomas often swell transiently for 6–18 months (pseudoprogression) and lose central enhancement before shrinking.
- Do not call failure on early growth alone; follow with serial MRI, and consider treatment failure only with sustained growth over about 2–3 years.
- Late effects to look for: hearing loss, trigeminal or facial symptoms, hydrocephalus and, rarely, radiation-induced malignancy.

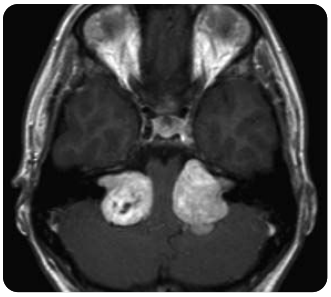


MRI, T1-weighted with gadolinium enhancement, axial. In an attempt to preserve residual hearing, the patient opted for fractionated stereotactic…

INNER EAR & IAC · TUMOURS

Neurofibromatosis II (NF II)

- Neurofibromatosis may initially present as otologic deficits. –
- The only ENT finding in cases of von Hippel–Lindau disease is the presence of a unilateral or bilateral endolymphatic sac tumor.
- Inflammatory diseases often have a rapidly progressive clinical course with multiple cranial nerve deficits, sometimes in combination with meningitis.



MRI, T1-weighted with gadolinium enhancement, axial. Bilateral CPA lesions, typical of schwannoma. The left mass has an anterior extension…

INNER EAR & IAC · TUMOURS

Meningioma of the Cerebellopontine Angle

IMAGING FINDINGS

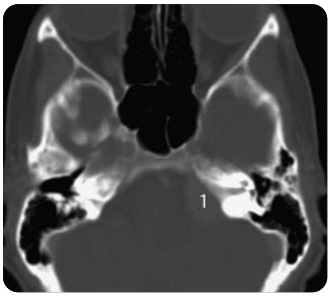
- Broad-based dural mass, obtuse angle with the petrous bone
- Eccentric to the porus; dural tail
- Calcification and hyperostosis; uniform enhancement

TOP DIFFERENTIALS

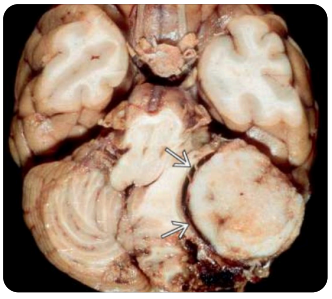
- Vestibular schwannoma
- Epidermoid cyst
- Dural metastasis

CHECKLIST

- Extension into the IAC, jugular foramen and Meckel cave
- Venous sinus involvement



CT, axial. Just before the birth of her child, screening with CT without contrast enhancement suggested a lesion in the left CPA (1). The size of…



Gross pathologic section viewed from below shows a large CPA meningioma with a broad dural base compressing the cerebellum. The specimen…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1228.

INNER EAR & IAC · INFECTION & INFLAMMATION

(Epi)dermoid Cyst

IMAGING FINDINGS

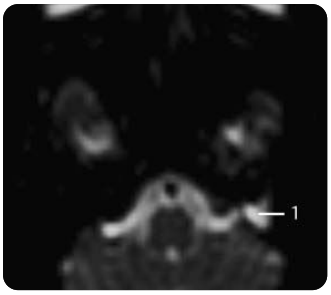
- CSF-like lesion that insinuates around nerves and vessels
- Incomplete suppression on FLAIR
- Bright on diffusion; no enhancement

TOP DIFFERENTIALS

- Arachnoid cyst (follows CSF on every sequence, DWI dark)
- Cystic schwannoma

CHECKLIST

- Encasement of cranial nerves and vessels
- Extent across cisterns before surgery



Patient with progressive and asymmetrical perceptive hearing loss.
Dunnebier, Imaging for Otolaryngologists, Fig. 5.9 a



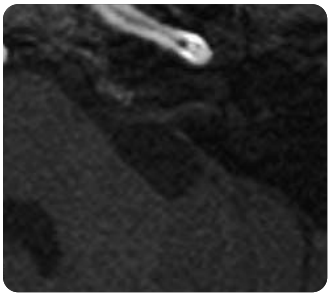
Axial CECT shows a large CPA epidermoid cyst. Note that this non-enhancing low-density lesion appears to invade the left cerebellar…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1202.

INNER EAR & IAC · INFECTION & INFLAMMATION

Arachnoid Cysts

- Epidermoid cysts, which are also hypointense on T1-weighted sequences and hyperintense on T2-weighted sequences, can be differentiated with diffusion-weighted imaging (DWI).
- In contrast, lipomas are hyperintense on T1-weighted images and hypointense on spin-echo T2-weighted sequences.
- Cysticercosis is seen in endemic areas, and the lesions are often smaller in size.



Patients evaluated for complaints of disabling tinnitus.
Dunnebier, Imaging for Otolaryngologists, Fig. 5.8 a